# Quizrr Chapter-Wise Test for JEE Main - 2025

**Q.1**

The value of $7 \log \frac{16}{15}+5 \log \frac{25}{24}+3 \log \frac{81}{80}$, is

**(A)** $\log 2$

**(B)** $\log 3$

**(C)** $\log 5$

**(D)** none of these

**Q.2**

The value of $\frac{\log 49 \sqrt{7}+\log 25 \sqrt{5}-\log 4 \sqrt{2}}{\log 17.5}$ is

**(A)** 5

**(B)** 2

**(C)** $\frac{5}{2}$

**(D)** $\frac{3}{2}$

**Q.3**

The value of $5^{\sqrt{\log_5 7}}-7^{\sqrt{\log_7 5}}$ is

**(A)** $\log 2$

**(B)** 1

**(C)** 0

**(D)** none of these

**Q.4**

The value of $2^{\log_3 7}-7^{\log_3 2}$ is

**(A)** $\log 2$

**(B)** 1

**(C)** 0

**(D)** none of these

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**Q.5**

If $9a^2 + 4b^2 = 18ab$, then $\log(3a + 2b) =$

**(A)** $\log 5 + \log 3 + \log a + \log 5b$

**(B)** $\log 5 + \log 3 + \log 3a + \log b$

**(C)** $\log 5 + \log a + \log b$

**(D)** none of these

**Q.6**

The value of $\log_{\sqrt{2}} \sqrt{2\sqrt{2\sqrt{2\sqrt{2}}}}$, is

**(A)** $\frac{15}{16}$

**(B)** $\frac{7}{16}$

**(C)** $\frac{15}{8}$

**(D)** $\frac{31}{32}$

**Q.7**

The equation $\log_e x + \log_e(1 + x) = 0$ can be written as

**(A)** $x^2 + x - 1 = 0$

**(B)** $x^2 + x + 1 = 0$

**(C)** $x^2 + x - e = 0$

**(D)** $x^2 + x + e = 0$

**Q.8**

The number of real values of the parameter $k$ for which the equation
$(\log_{16} x)^2 - \log_{16} x + \log_{16} k = 0$
with real coefficients will have exactly one solution, is

**(A)** 2

**(B)** 1

**(C)** 4

**(D)** none of these

**Q.9**

If $x = \log_a(bc), y = \log_b(ca)$ and $z = \log_c(ab)$, then which of the following is correct?

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**(A)** $x+y+z=1$

**(B)** $\frac{1}{1+x}+\frac{1}{1+y}+\frac{1}{1+z}=1$

**(C)** $xyz=1$

**(D)** none of these

**Q.10**

If $\log_2 \sin x-\log_2 \cos x-\log_2\left(1-\tan^2 x\right)=-1$, then $x=$

**(A)** $\frac{n\pi}{2}+\frac{\pi}{8}, n\in Z$

**(B)** $n\pi-\frac{\pi}{8}, n\in Z$

**(C)** $\frac{n\pi}{4}+\frac{\pi}{2}, n\in Z$

**(D)** none of these

**Q.11**

If $\log_a x, \log_b x, \log_c x$ are in A.P., where $x\neq 1$, then $c^2=$

**(A)** $(ab)^{\log_b b}$

**(B)** $(ac)^{\log_a b}$

**(C)** $(ab)^{\log_b a}$

**(D)** $(ac)^{\log_b a}$

**Q.12**

If $2^{\log_{10} \sqrt[3]{3}}=3^{k\log_{10} 2}$, then $k=$

**(A)** $\frac{1}{2}$

**(B)** $\frac{3}{2}$

**(C)** 3

**(D)** 2

**Q.13**

If $\log_{10} 2=0.3010$, then $\log_5 64=$

**(A)** $\frac{602}{233}$

**(B)** $\frac{233}{602}$

**(C)** $\frac{202}{633}$

**(D)** $\frac{633}{202}$

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**Q.14**

The value of $(0.16)^{\log_{2.5}\left(\frac{1}{3}+\frac{1}{3^2}+\frac{1}{3^3}+\ldots+\infty\right)}$ is

**(A)** $0.16$

**(B)** $1$

**(C)** $0.4$

**(D)** $4$

**Q.15**

If $\log_{30} 3 = x, \log_{30} 5 = y$, then $\log_{30} 8 =$

**(A)** $3(1-x-y)$

**(B)** $x-y+1$

**(C)** $1-x-y$

**(D)** $2(x-y+1)$

## Answers & Solutions

**Q.1 Answer:**

$\log 2$

**Solution:**

$$
\begin{aligned}
&7\log\tfrac{16}{15}+5\log\tfrac{25}{24}+3\log\tfrac{81}{80}\\
&=\log\left(\tfrac{16}{15}\right)^7+\log\left(\tfrac{25}{24}\right)^5+\log\left(\tfrac{81}{80}\right)^3\\
&=\log\left(\left(\tfrac{16}{15}\right)^7\times\left(\tfrac{25}{24}\right)^5\times\left(\tfrac{81}{80}\right)^3\right)\\
&=\log\left(\tfrac{2^{28}}{3^7\times5^7}\times\tfrac{5^{10}}{2^{15}\times3^5}\times\tfrac{3^{12}}{2^{12}\times5^3}\right)\\
&=\log\left(2^{28-12-15}\times5^{10-7-3}\times3^{12-7-5}\right)\\
&=\log(2)
\end{aligned}
$$

**Q.2 Answer:**

$\frac{5}{2}$

**Solution:**

$$
\begin{aligned}
&=\tfrac{\log 49\sqrt{7}+\log 25\sqrt{5}-\log 4\sqrt{2}}{\log(17.5)}\\
&=\tfrac{\log 7^2\cdot7^{1/2}+\log 5^2\cdot5^{1/2}-\log 2^2\cdot2^{1/2}}{\log(17\cdot5)}\\
&=\tfrac{\log 7^{5/2}+\log 5^{5/2}-\log 2^{5/2}}{\log(17\cdot5)}\\
&=\tfrac{\frac{5}{2}(\log 7+\log 5-\log 2)}{\log(17-5)}\\
&=\tfrac{\frac{5}{2}\left(\log\left(\frac{7\times5}{2}\right)\right)}{\log(17-5)}
\end{aligned}
$$

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$= \frac{5}{2} \times \frac{\log(17-5)}{\log(17-5)}$
$= \frac{5}{2}$

**Q.3 Answer:**

0

**Solution:**

$5^{\sqrt{\log_5 7}} - 7^{\sqrt{\log_7 5}}$ **Put** $7 = 5^t$
$5^{\sqrt{\log_5 5^t}} - 7^{\sqrt{\log_5 t^5}}$
$5^{\sqrt{t}} - 7^{\sqrt{t}}$
$5^{\sqrt{t}} - (5^t)^{1/\sqrt{t}} = 5^{\sqrt{t}} - 5^{t/\sqrt{t}} = 5^{\sqrt{t}} - 5^{\sqrt{t}} = 0$

**Q.4 Answer:**

0

**Solution:**

$2^{\log_3 7} - 7^{\log_3 2}$
**Let** $2^{\log_3 7} = x$
**Taking** $\log$ **both sides (the base 3 )**
$\log_3 7 \cdot \log_3 2 = \log_3 x \ldots\ldots \left(2^{\log_3 7}\right) = \log_3 x$
**Let** $7^{\log_3 2} = y$
**Taking** $\log$ **both sides (the base 3)**
$\log_3 2 \cdot \log_3 7 = \log_3 y \ldots\ldots\ldots (2)$
**From eq.** $(1)$ **and** $(2)$
**Both are equal it means** $x = y$
**Answer** $= 0$

**Q.5 Answer:**

none of these

**Solution:**

$9a^2 + 4b^2 = 18ab$
$(3a + 2b)^2 = 9a^2 + 4b^2 + 12ab$
$(3a + 2b)^2 - 12ab = 18ab$
$(3a + 2b)^2 - 12ab = 9a^2 + 4b^2$
$(3a + 2b)^2 = 30ab$
$3a + 2b = (30ab)^{1/2}$
$\log(3a + 2b) = \log(30ab)^{1/2} = \frac{1}{2}\log(30ab)$

**Q.6 Answer:**

$\frac{15}{8}$

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**Solution:**

$$\log_{\sqrt{2}} \sqrt{2\sqrt{2\sqrt{2\sqrt{2}}}}$$
$$= \tfrac{2}{2} \log_2 2\sqrt{2\sqrt{2\sqrt{2}}}$$
$$= \log_2 2\sqrt{2\sqrt{2\sqrt{2}}}$$
$$= \log_2 2 + \log_2 \sqrt{2\sqrt{2\sqrt{2}}}$$
$$= 1 + \tfrac{1}{2} \log_2 2\sqrt{2\sqrt{2}}$$
$$= 1 + \tfrac{1}{2} \left[\log_2 2 + \log_2 \sqrt{2\sqrt{2}}\right]$$
$$= 1 + \tfrac{1}{2} + \tfrac{1}{2} \left[\tfrac{1}{2} \log_2 2\sqrt{2}\right]$$
$$= \tfrac{3}{2} + \tfrac{1}{2} \left[\tfrac{1}{2} \left(\log_2 2 + \log_2 \sqrt{2}\right)\right]$$
$$= \tfrac{3}{2} + \tfrac{1}{2} \left[\tfrac{1}{2} \left(1 + \tfrac{1}{2} \log_2 2\right)\right]$$
$$= \tfrac{3}{2} + \tfrac{1}{2} \times \tfrac{1}{2} \left(1 + \tfrac{1}{2}\right)$$
$$= \tfrac{3}{2} + \tfrac{1}{4} \times \tfrac{3}{2}$$
$$= \tfrac{3}{2} \times \tfrac{5}{4}$$
$$= \tfrac{15}{8}$$

**Q.7 Answer:**

$x^2 + x - 1 = 0$

**Solution:**

$$\log_c x + \log_c (1 + x) = 0$$
$$\log_e x(1 + x) = 0$$
$$x(x + 1) = 1$$
$$x^2 + x = 1$$
$$x^2 + x - 1 = 0$$

**Q.8 Answer:**

1

**Solution:**

**Let** $\log_{16} x = y$

**Then, given equation becomes,** $y^2 - y + \log_{16} k = 0$

**This is a quadratic equation and it has exactly one solution.**

$$\therefore D = 0$$
$$\Rightarrow b^2 - 4ac = 0$$
$$\Rightarrow (-1)^2 - 4(1)\left(\log_{16} k\right) = 0$$
$$\Rightarrow \log_{16} k = \tfrac{1}{4}$$
$$\Rightarrow k = 16^{\frac{1}{4}}$$
$$\Rightarrow k = 2$$

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**So, there is only one value for $k$.**

**So, option $(b)$ is the correct option.**

**Q.9 Answer:**

$\frac{1}{1+x}+\frac{1}{1+y}+\frac{1}{1+z}=1$

**Solution:**

$x=\log_a bc$

$x=\frac{\log bc}{\log 4}$

$x=\frac{\log b+\log c}{\log a}$

$x+1=\frac{\log b+\log c}{\log a}+1$

$x+1=\frac{\log b+\log c+\log a}{\log a}$

$\frac{1}{x+1}=\frac{\log a}{\text{tog } a+\log b+\log c}$

**similarly** $\frac{1}{y+1}=\frac{\log b}{\log a+\log b+\log c}$

\& $\frac{1}{z+1}=\frac{\text{tog } c}{\log a+\log b+\log c}$

$\frac{1}{x+1}+\frac{1}{y+1}+\frac{1}{2+1}=\frac{\log a}{\log a+\log b+\log c}+\frac{\log b}{\log a+\log b+\log c}+\frac{\log c}{\log a+\log b+\log c}$

**Q.10 Answer:**

$\frac{n\pi}{2}+\frac{\pi}{8}, n\in Z$

**Solution:**

**Given:** $\log_2 \sin x-\log_2 \cos x-\log_2\left(1-\tan^2 x\right)=-1$

$\therefore \log_c(a)-\log_c b=\log_c\left(\frac{a}{b}\right)=-1$

$\Rightarrow \log_2\left(\frac{\sin x}{\cos x}\right)-\log_2\left(1-\tan^2 x\right)=-1$

$\Rightarrow \log_2\left(\frac{\tan(x)}{1-\tan^2 x}\right)=-1$

$\Rightarrow \log_2\left(\frac{\tan(x)}{1-\tan^2 x}\right)=\log_2\left(\frac{1}{2}\right)$

$\Rightarrow \frac{\tan(x)}{1-\tan^2 x}=\frac{1}{2}$

$\Rightarrow \frac{2\tan x}{1-\tan^2 x}=1$

$\Rightarrow \tan 2x=1$

$\Rightarrow \tan 2x=n\pi+\frac{\pi}{4}$

$\Rightarrow x=\frac{n\pi}{2}+\frac{\pi}{8}$

**Q.11 Answer:**

$(ac)^{\log_a b}$

**Solution:**

$\log_a x, \log_b x, \log_c x$ **AP**

$2\log_b x=\log_a x+\log c^x$

$\Rightarrow 2\left(\frac{\log x}{\log b}\right)=\frac{\log x}{\log a}+\frac{\log x}{\log c}$

$\frac{2}{\log b}=\frac{\log a+\log c}{\log a\log c}$

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$2\log c = \left(\frac{\log b}{\log a}\right)(\log c + \log a)$

$2\log c = \log_a b(\log ac)$

$\log c^2 = log(ac)^{\log_a b}$

$c^2 = (ac)^{\log_a b}$

**Q.12 Answer:**

$\frac{3}{2}$

**Solution:**

$2^{\log_{10} 3\sqrt{3}} = 3^{k\log_{10} 2} \Rightarrow 2^{\log_{10}\left(3^{\frac{3}{2}}\right)} = 3^{k\log_{10} 2}$

$\Rightarrow 2^{\log_2\left(3^{\frac{3}{2}}\right)\cdot\log_{10} 2} = 3^{k\log_{10} 2} \quad [\text{Using } \log_a x = \log_b x \cdot \log_a b]$

$\Rightarrow \left[2^{\log_2\left(3^{\frac{3}{2}}\right)}\right]^{\log_{10}2} = \left(3^k\right)^{\log_{10} 2} \Rightarrow 2^{\log_2 3^{\frac{3}{2}}} = 3^k$

$\Rightarrow 3^{\frac{3}{2}} = 3^k \Rightarrow k = \frac{3}{2} \quad \left(\because a^{\log_a x} = x\right)$

**Q.13 Answer:**

$\frac{602}{233}$

**Solution:**

$\log_5 64 = \log_5 2^6$

$= 6\log_5 2 \quad ...(1)$

$\log_{10} 2 = 0.3010$

$\frac{\log_5 2}{\log_5 10} = 0.3010$

$\log_5 2 = 0.3010$

$\log_5(5 \times 2)$

$\log_5 2$

$\log_5^5 + \log_5^2$

$\log_5^2 = t$

$\frac{t}{1+t} = 0.3010$

$t = 0.3010 + 03010t$

$\Rightarrow t = \frac{0.3010}{0.6990}$

$\log_5 64 = 6 \times \log_5 2$

$= 6 \times \frac{0.3010}{0.6990}$

$\log_5 64 = \frac{602}{233}$

**Correct option (1)**

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**Q.14 Answer:**

4

**Solution:**

**Before solving this question, you have to be acquainted with certain identities:**

$a^{[\log b(\text{base } a)]} = b \quad \ldots(1)$

$a^{[\log b(\text{base } a^n)]} = b^{(\frac{1}{n})} \quad \ldots(2)$

**sum of infinite G.P,** $= a/(1-r) \ldots (3)$

**Solving the exponent part:**

$1/3 + (1/3)^2 + \ldots\ldots\ldots\ldots\ldots (1/3)^{\infty} = (1/3)/[1 - 1/3] = 1/2 \ldots$ **using(3)**

$0.16 = 4/25$

$2.5 = 5/2$

$= \left(\frac{4}{25}\right)^{\log_{(25)}\left(\frac{1}{2}\right)}$

$= \left(\frac{1}{2}\right)^{\log_{(25)}\left(\frac{4}{25}\right)}$

$= \left(\frac{1}{2}\right)^{-2} = 4$

**Q.15 Answer:**

$3(1 - x - y)$

**Solution:**

$\log_{30} 3 = a$ **and** $\log_{30} 5 = b$

**To find** $\log_{30} 8$**, we will use** $\log_{30} 30 = 1$

$\implies \log_{30} 30 = 1$

$\implies \log_{30}(3 \times 5 \times 2) = 1$

$\implies \log_{30} 3 + \log_{30} 5 + \log_{30} 2 = 1$

$\implies \mathbf{a} + \mathbf{b} + \log_{30} 2 = 1$

$\implies \log_{30} 2 = (1 - a - b)$

$\therefore \log_{30} 8 = \log_{30} 2^3 = 3 \log_{30} 2 = 3(1 - \mathrm{a} - \mathrm{b})$

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